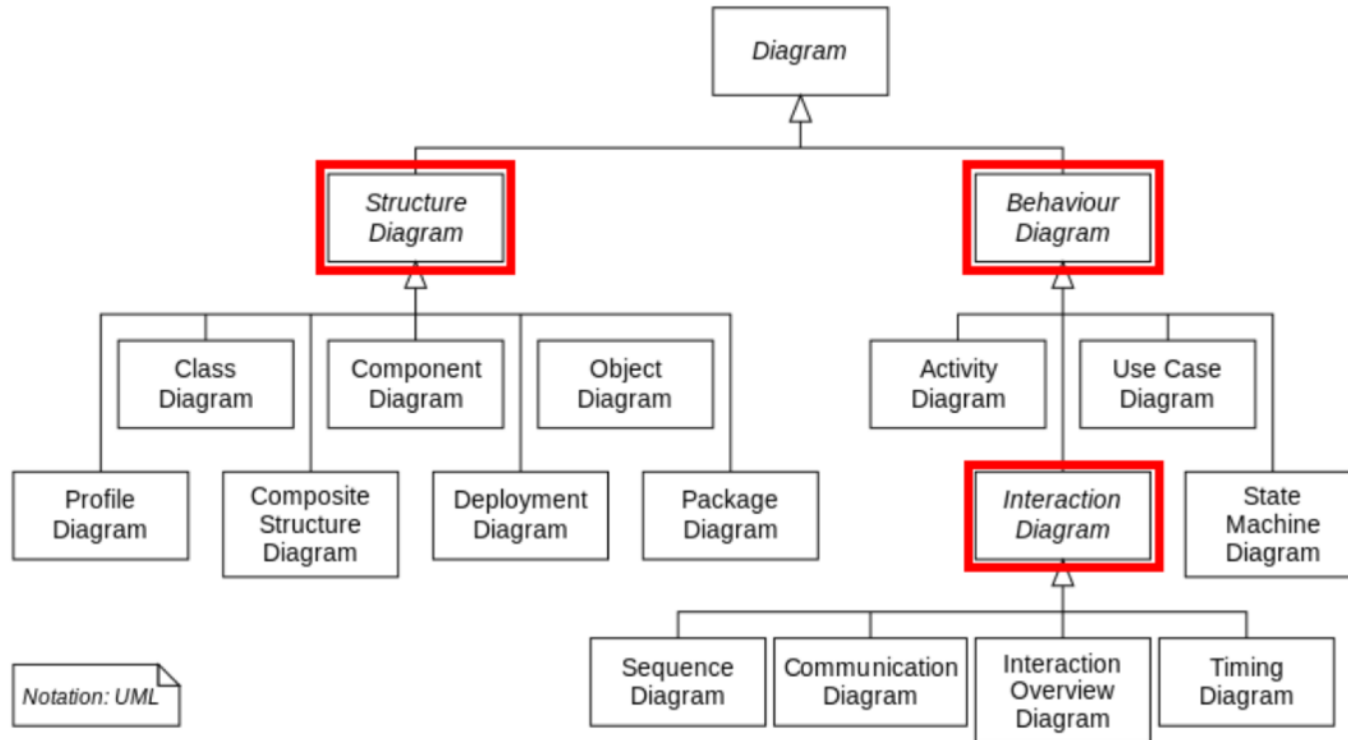


UML User and Activity Diagrams

A.Y. 2020/2021

UML diagram taxonomy



Use case diagram

A representation of a user's interaction with the system that shows the relationship between the user and the different use cases in which the user is involved.

A use case diagram can identify the different types of users of a system and the different use cases and will often be accompanied by other types of diagrams as well.

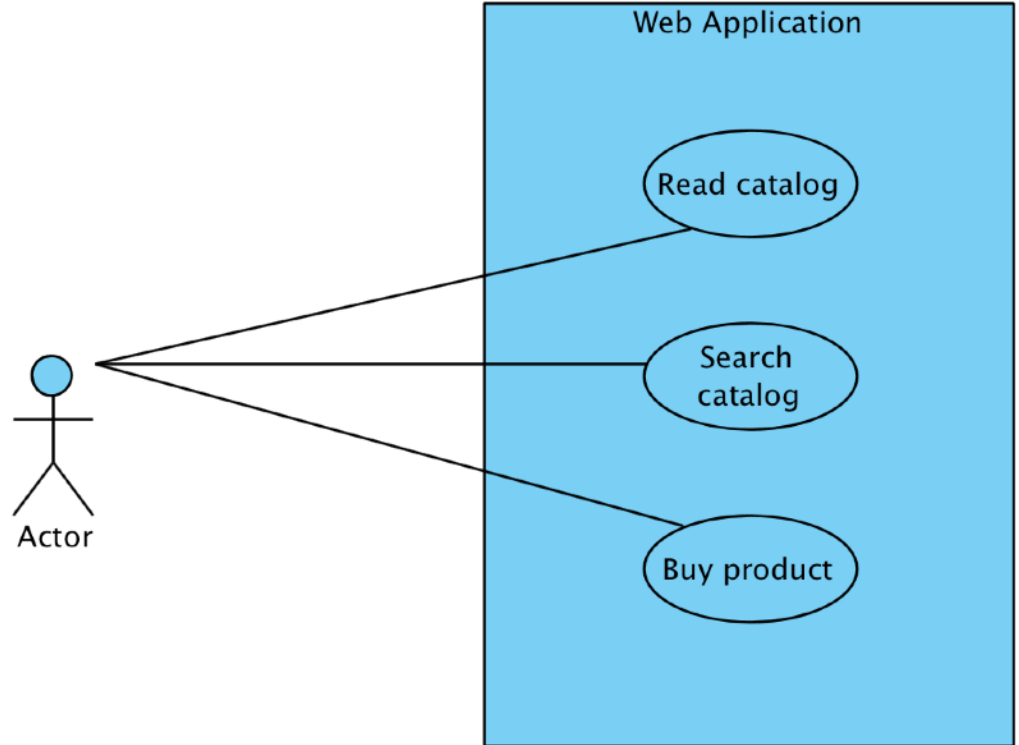
Use case diagram: basic notation



Associations

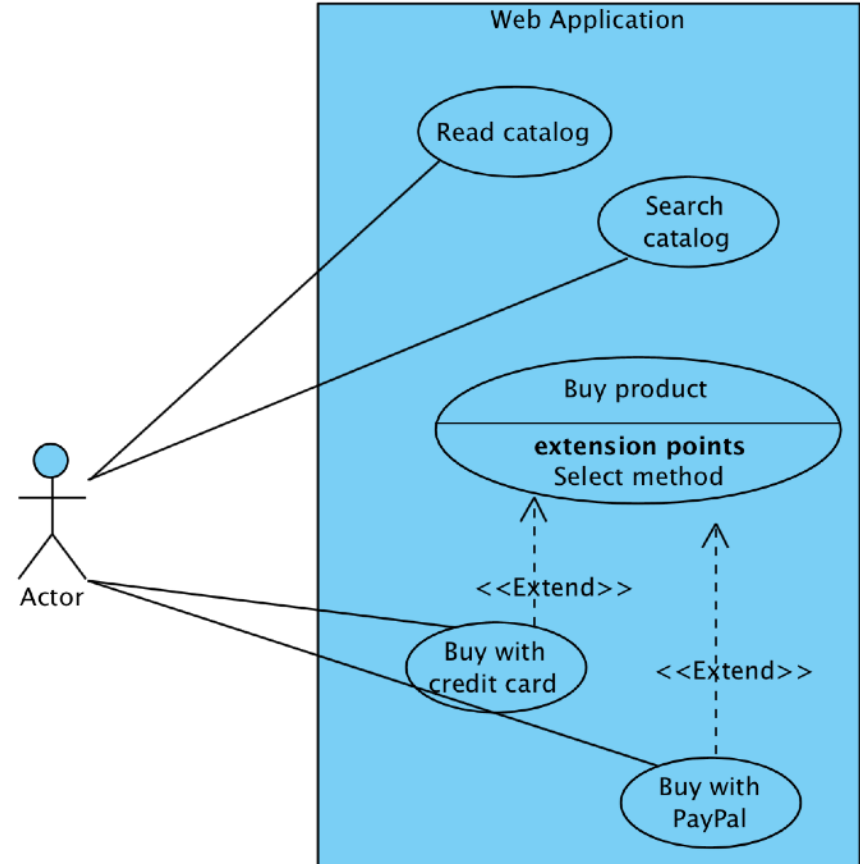
A web application allows a user to

- read the products in the catalog
- search products in the catalog
- buy a product.



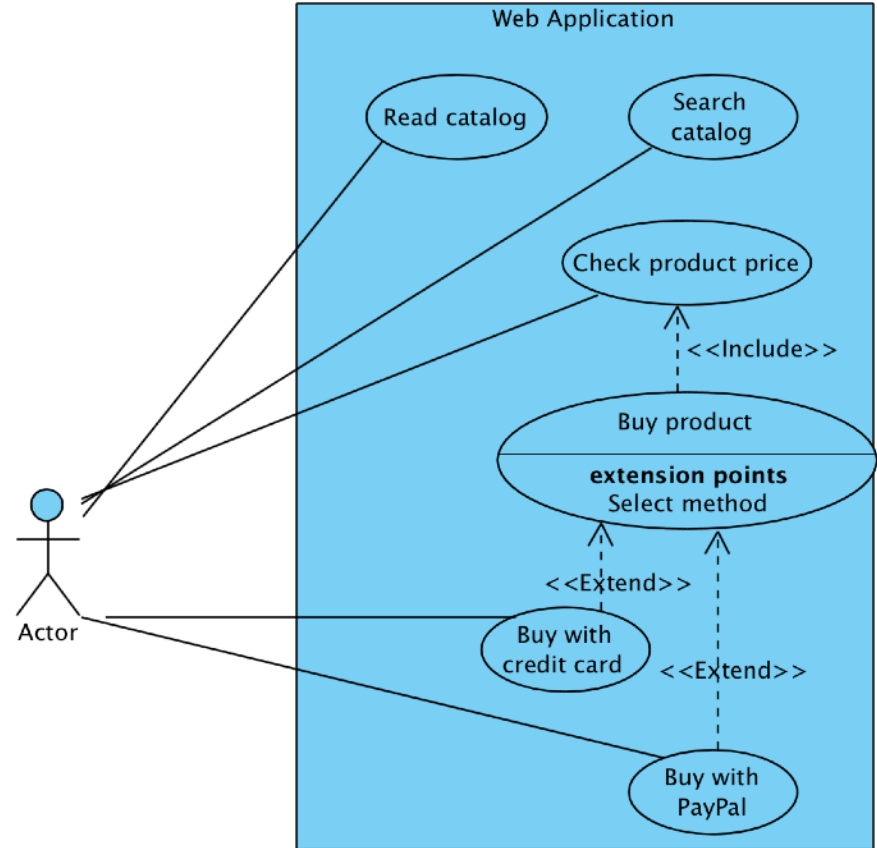
Extension points

An extension point is a feature of a use case which identifies (references) a point in the behaviour of the use case where that behaviour can be extended by some other (extending) use case, as specified by extend relationship.



Includes

A directed relationship between two use cases which is used to show that behaviour of the included use case (the addition) is inserted into the behaviour of the including (the base) use case.



Use Case Specification

Just showing the use case diagram in UML notation is not enough. Each use case accompanied by text explaining the purpose of the use case as well as what functionality is accomplished when a use case is executed.

Provides detailed information about a use case with:

- Actors
- Preconditions
- Flow of events (main and alternatives)
- Postconditions

Example

Use case: Buy with credit card
ID: UC1
Actors: User
Preconditions: 1. The shopping cart contains some item 2. The user select credit card as method for paying
Flow of events: 1. The system calculate the bill 2. The system shows the bill to the user 3. The user provides credit card details 4. The system records credit card details 5. The system notifies the user about the successful operation
Postconditions: 1. The user receives a message that confirms the purchase 2. The system issues the receipt

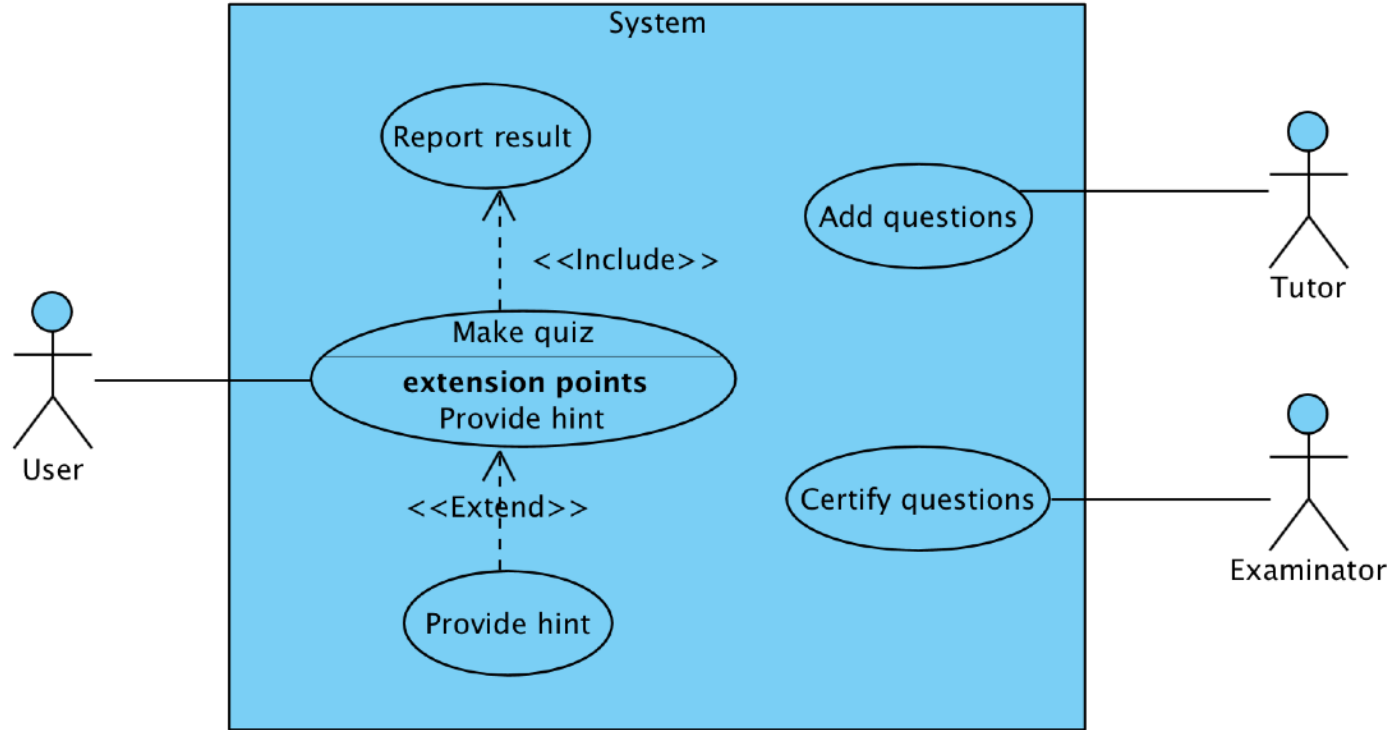
Exercise

A user can request a quiz for the system. The system picks a set of questions from its database, and compose them together to make a quiz. It rates the user's answers, and gives hints if the user requests it.

In addition to users, we also have tutors who provide questions and hints. And also examiners who must certify questions to make sure they are not too trivial, and that they are sensical.

Make a use case diagram to model this system. Work out some of your use cases.

Solution



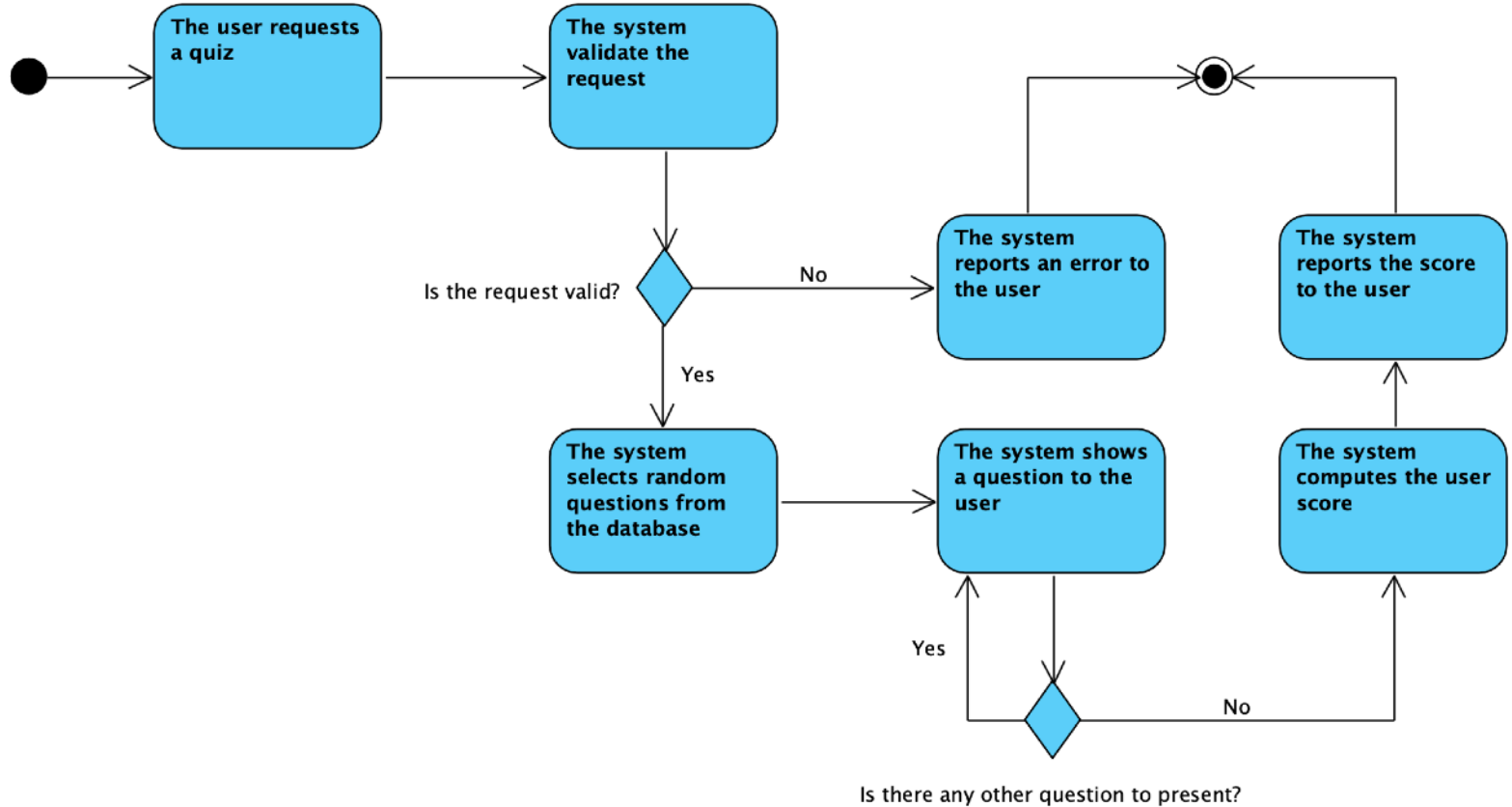
Activity diagrams

An activity diagram is basically a flowchart to represent the flow from one activity to another activity. The activity can be described as an operation of the system.

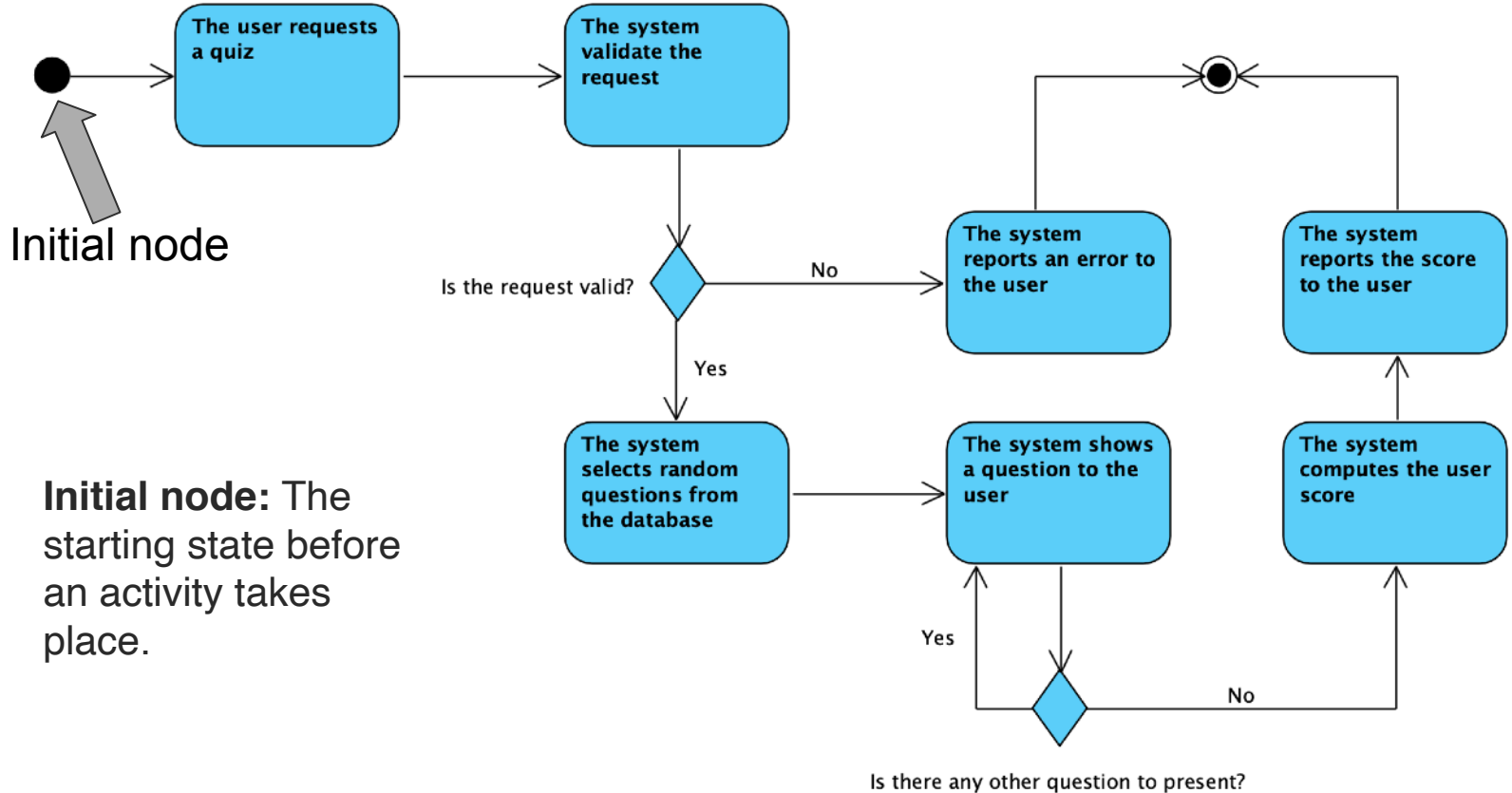
The control flow is drawn from one operation to another. This flow can be sequential, branched, or concurrent. Activity diagrams deal with all type of flow control by using different elements such as fork, join, etc

Activity diagrams capture the dynamic behavior of the system.

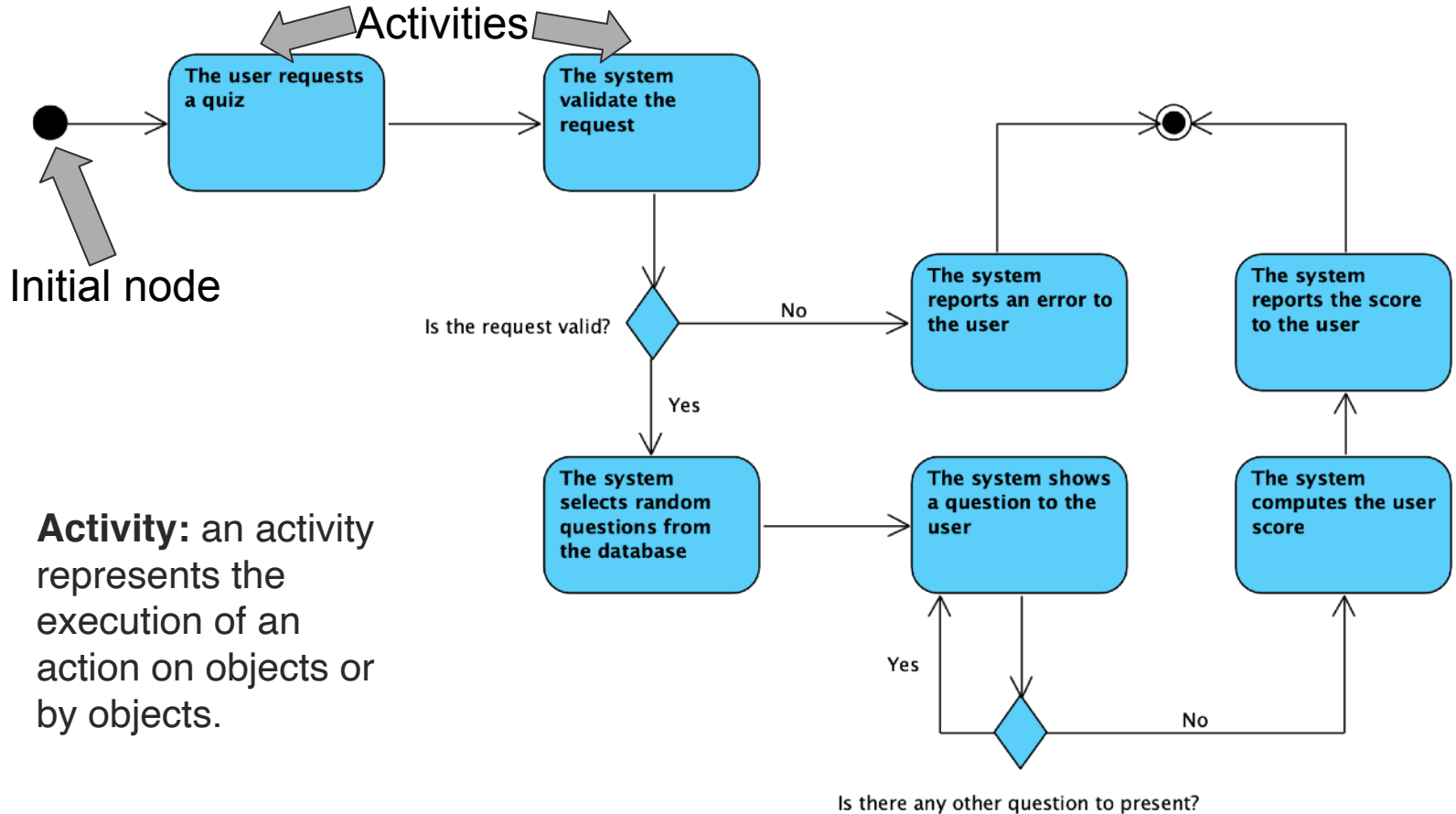
Notation by examples



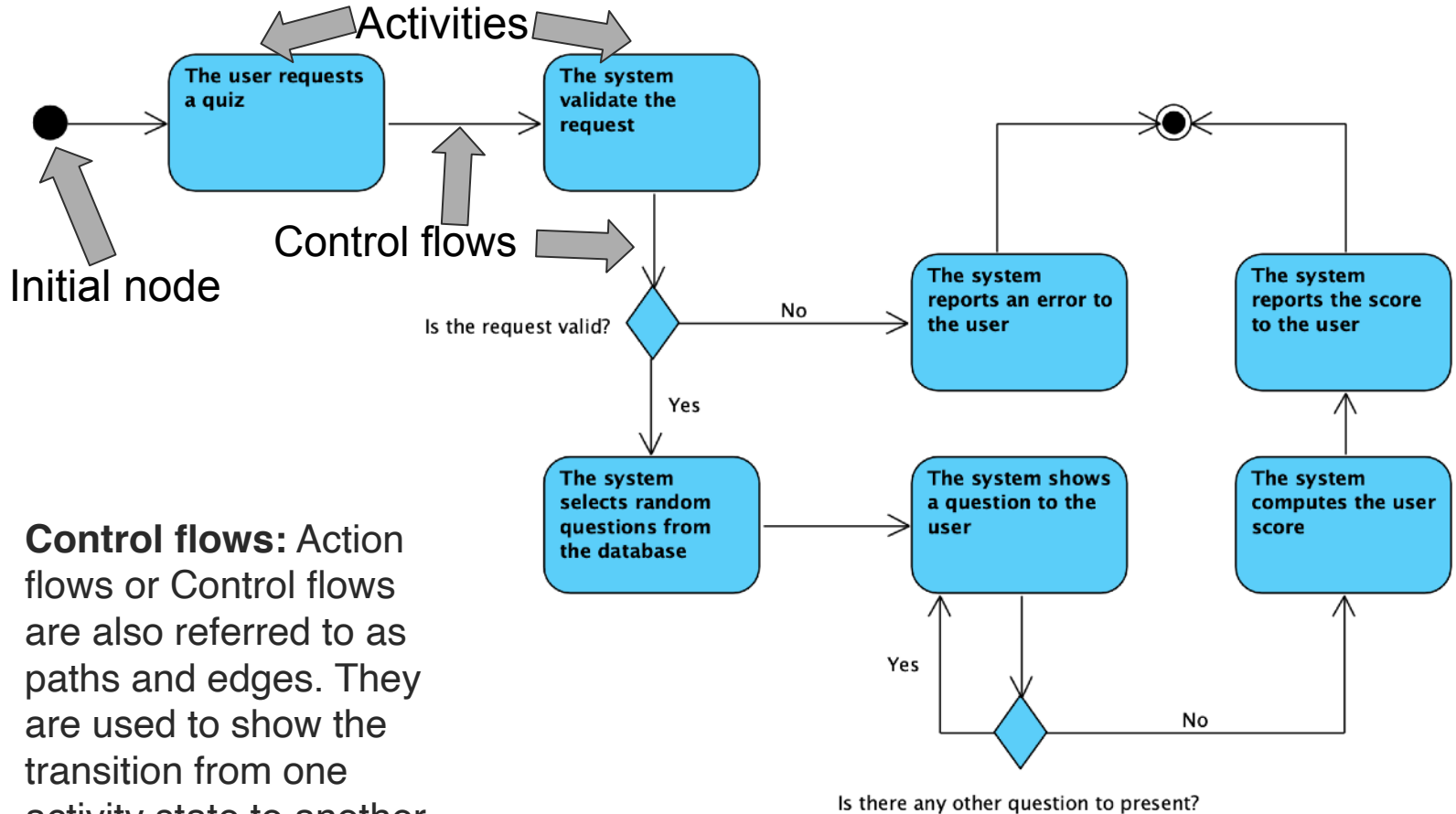
Notation by examples



Notation by examples

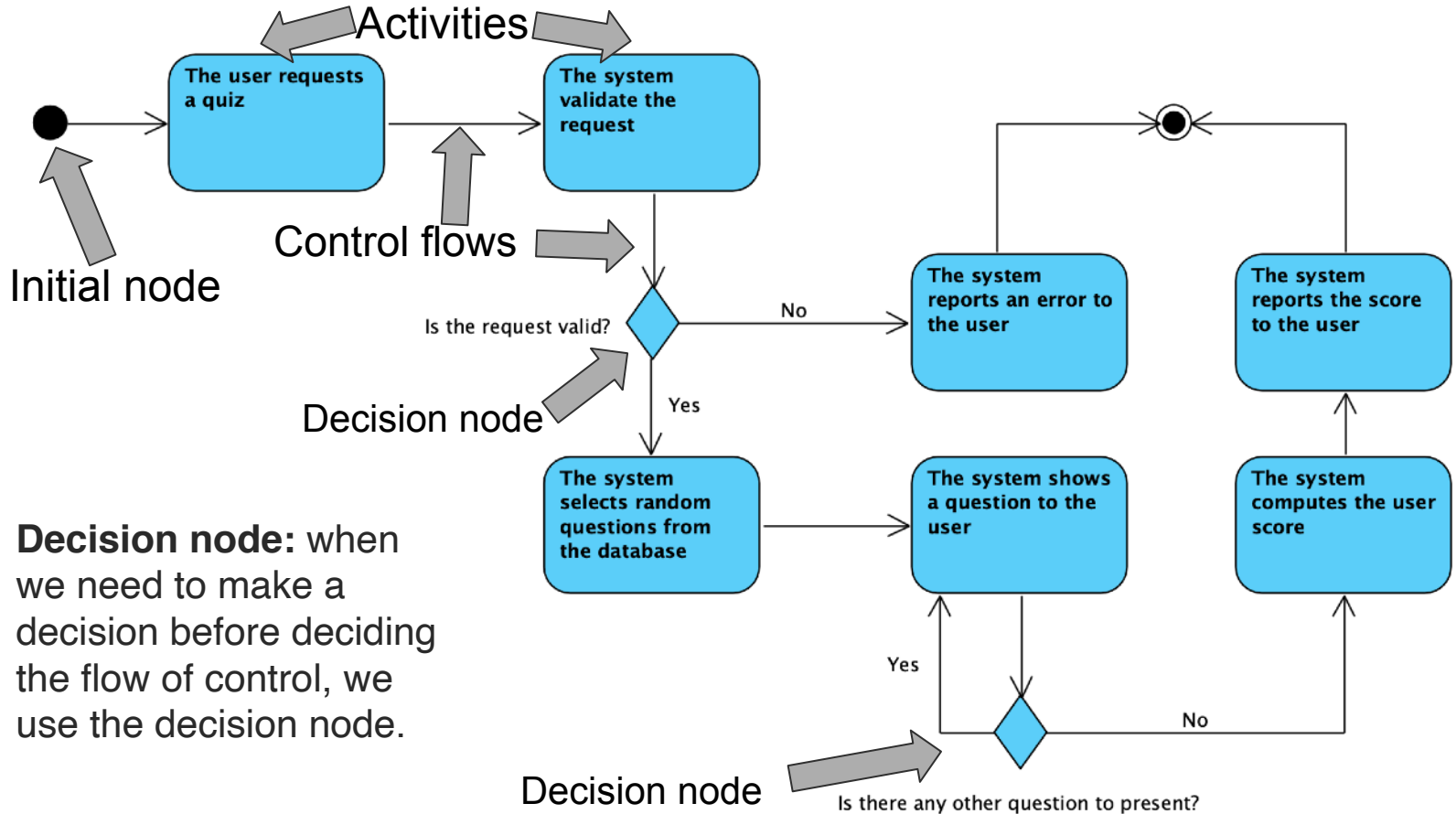


Notation by examples

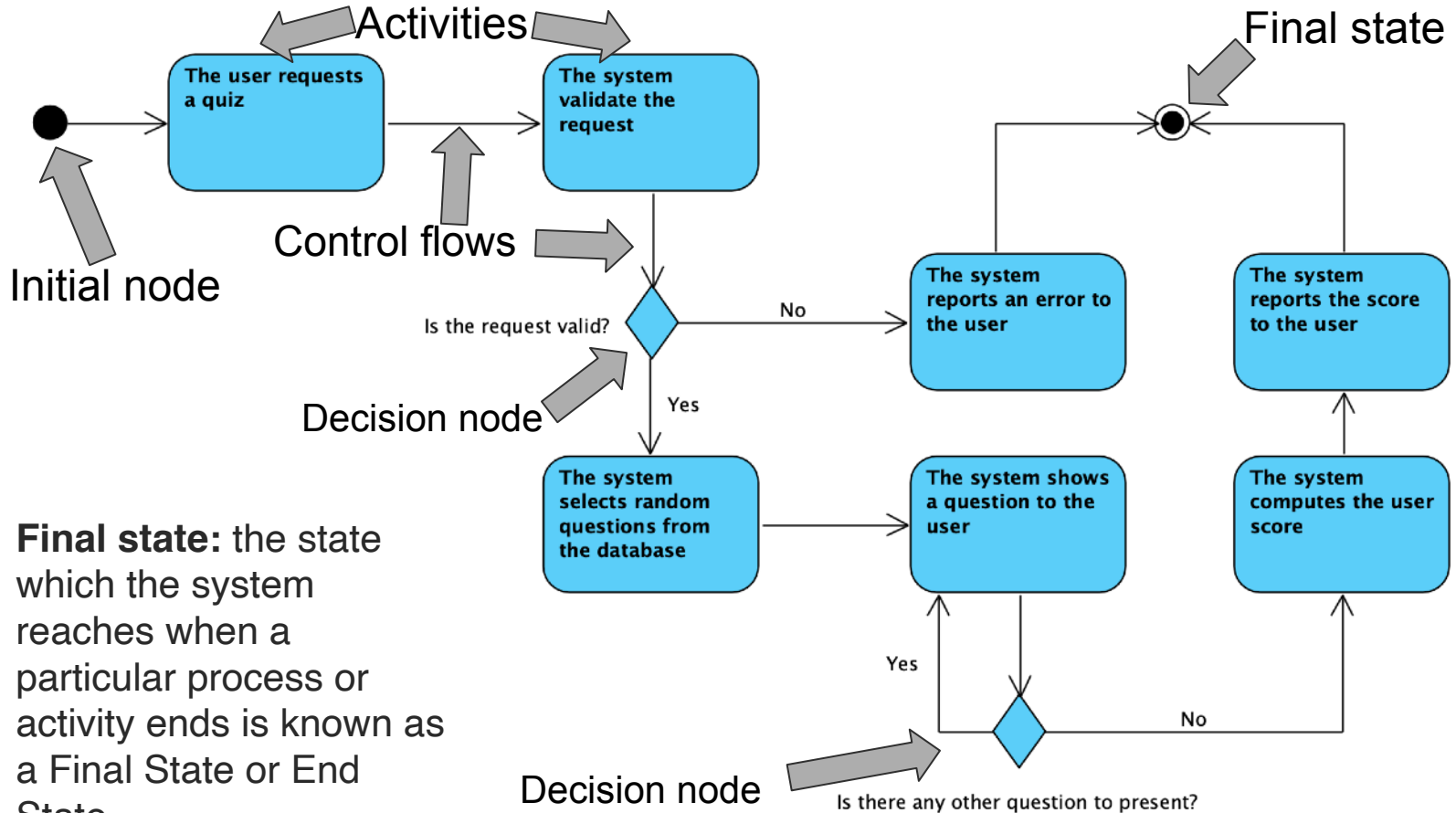


Control flows: Action flows or Control flows are also referred to as paths and edges. They are used to show the transition from one activity state to another.

Notation by examples



Notation by examples



Exercise

Model the activity diagram associated with the use case

Use case: Buy with credit card
ID: UC1
Actors: User
Preconditions: 1. The shopping cart contains some item 2. The user select credit card as method for paying
Flow of events: 1. The system calculate the bill 2. The system shows the bill to the user 3. The user provides credit card details 4. The system records credit card details 5. The system notifies the user about the successful operation
Postconditions: 1. The user receives a message that confirms the purchase 2. The system issues the receipt